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Paper 1: Population Pharmacokinetics of Serplulimab and Quantitative Assessment of Transi

Bibliographic Information

- **Title**: Population Pharmacokinetics of Serplulimab and Quantitative Assessment of Transitioning From Weight-Based to Flat-Dosing Strategy
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Quick Take

This comprehensive PopPK analysis of serplulimab (anti-PD-1 mAb) in $2,110$ patients across 11 trials provides robust quantitative justification for transitioning from weight-based (3 mg/kg Q2W, 4.5 mg/kg Q3W) to flat-dose (200 mg Q2W, 300 mg Q3W) regimens. Essential reading for pharmacometricians supporting regulatory submissions for oncology biologics, particularly those navigating FDA/EMA criteria for dose regimen changes. The integrated E-R analyses and cross-tumor-type modeling approach set a methodological benchmark for flat-dose justification.

Executive Summary

Wang et al. present a methodologically rigorous, large-scale PopPK analysis that successfully bridges serplulimab's development from weight-based to flat-dosing, a transition increasingly sought for monoclonal antibodies to simplify clinical workflow. Using $2,110$ patients from 11 trials across six tumor types, they developed a two-compartment model with time-varying clearance, identified modest covariate effects ($\leq 20\%$), and demonstrated through simulation that flat-dose regimens achieve exposures within FDA-recommended comparability bounds ($< 20\%$ difference in AUC/C_{min}, $< 25\%$ in C_{max}). Critically, exposure-response analyses confirmed no meaningful efficacy or safety differences, providing a template for quantitative dose regimen optimization that aligns with evolving regulatory expectations for PD-1/PD-L1 inhibitors.

⚡ Methodological Snapshot

The authors employed a two-compartment population PK model with time-varying clearance, implemented in NONMEM (FOCEI or SAEM), to characterize serplulimab disposition across $2,110$ patients from 11 clinical trials spanning Phase I-III. The structural model incorporated an empirical time-varying clearance component (sigmoid Emax function of time) rather than target-mediated drug disposition (TMDD), as PD-1 receptor saturation occurs at low concentrations, precluding reliable TMDD parameter estimation. Covariate analysis followed a stepwise forward-addition/backward-elimification procedure, with tumor type-specific parameter estimates to address potential heterogeneity across indications. Model evaluation included bootstrap resampling ($n = 1,000$), prediction-corrected visual predictive checks stratified by tumor type and

dose, and extensive simulation-based sensitivity analyses for covariate effects and dosing regimen comparisons.

⌧⌧ Structural Model Breakdown

The final model comprises two compartments (central V_c, peripheral V_p) with first-order inter-compartmental clearance Q and time-varying systemic clearance $CL(t)$. Baseline clearance CL_0 varies by tumor type (7 categories: Sq. NSCLC, HCC, CRC, Non-sq. NSCLC, SCLC, ESCC, Other) with allometric scaling on weight (power 0.514), albumin effect (power -0.714), tumor burden (power 0.0548), ALP (power 0.0553), and sex (female -14.5% lower clearance). Central volume V_c is tumor type-specific with weight scaling (power 0.470), albumin effect (power -0.320), and sex effect (female -14%). Peripheral volume $V_p = 2.98$ L with albumin (power -1.05) and tumor burden (power 0.107) effects. Inter-compartmental clearance $Q = 0.0169$ L/day with IIV only. Time-varying clearance follows sigmoid Emax model with $T_{50} = 221$ days, Hill coefficient $\lambda = 2.43$, and $E_{max} = -0.0926$ (9.26% reduction from baseline), capturing disease-related clearance changes over treatment course.

Detailed Methodological Analysis

Modeling Approach

Two-compartment population PK model with first-order elimination and time-varying clearance, implemented in NONMEM. Time-varying clearance modeled via sigmoid Emax function: $CL(t) = CL_0 \times \exp\left[E_{max} \times (time/T_{50})^{\lambda} / \left(1 + (time/T_{50})^{\lambda}\right) + \eta_{CL}\right]$. Alternative TMDD structures (simplified Michaelis-Menten) were explored but rejected due to rapid target saturation limiting informative data. Tumor type-specific baseline clearance (CL_0) and central volume (V_c) parameters estimated separately for Sq. NSCLC, HCC, CRC, Non-sq. NSCLC, SCLC, ESCC, and Other to preserve indication-specific applicability.

Data Sources

Pooled dataset from 11 clinical trials: 8 previously reported trials (2 Phase I, 4 Phase II, 2 Phase III) with updated cutoff dates plus 3 new Phase III trials (HLX10-002-NSCLC301, HLX10-007-EC301, HLX10-015-mCRC301). Total $2,110$ patients with serplulimab concentration-time data across hepatocellular carcinoma, colorectal cancer, squamous and non-squamous NSCLC, small cell lung cancer, and esophageal squamous cell carcinoma. Rich and sparse PK sampling with median baseline weight 62 kg (range $32.9 - 131$ kg), 79% male, 87% Asian.

Estimation Methods

Nonlinear mixed-effects modeling in NONMEM with FOCEI or SAEM estimation. Inter-individual variability (IIV) modeled as exponential (log-normal) distributions for CL, V_c, Q, V_p, and E_{max}. Covariance between CL and V_c IIV estimated. Residual variability as proportional error model. Final parameter precision assessed via non-parametric bootstrap ($1,000$ replicates).

Model Evaluation

Comprehensive evaluation including: (1) standard diagnostic plots (observed vs. predicted, conditional weighted residuals); (2) prediction-corrected visual predictive checks (pcVPCs) stratified by tumor type and dose regimen; (3) bootstrap analysis for parameter uncertainty and bias assessment; (4) sensitivity analyses for covariate impact on exposure; (5) eta shrinkage assessment (CL 14.9%, V_c 27.0%, Q 67.3%, V_p 35.8%, E_{max} 46.7%). Tumor type-specific pcVPCs confirmed adequate model performance across all indications.

Covariate Analysis

Stepwise forward-addition ($p < 0.05$) and backward-elimination ($p < 0.01$) procedure on baseline covariates: body weight, albumin, sex, tumor burden, alkaline phosphatase. Continuous covariates normalized to population median and modeled as power functions; categorical covariates as exponential fractional changes. Exploratory subgroup analyses (not in formal model) for age, race, hepatic/renal function, ADA status using empirical Bayes estimates (EBE) for exposure simulation. Covariate effects classified: not meaningful ($\leq 20\%$), minor ($20 - 30\%$), moderate/large ($> 30\%$).

Scientific Context & Motivation

The transition from weight-based to flat-dosing for therapeutic antibodies represents a paradigm shift in oncology drug development, driven by operational efficiency and reduced medication errors, but requires rigorous pharmacometric justification to ensure maintained benefit-risk profiles. For PD-1/PD-L1 inhibitors specifically, flat-dosing has been adopted for several agents (pembrolizumab, nivolumab), yet each antibody requires individualized assessment due to differences in target binding, FcRn recycling, and immunogenicity. This study addresses a critical knowledge gap by providing the first comprehensive, multi-tumor-type PopPK characterization of serplulimab with explicit quantitative criteria for dose regimen transition. The work builds upon prior serplulimab PK modeling ($1,144$ patients) and aligns with FDA's 2022 guidance on PK-based criteria for alternative dosing regimens of PD-1/PD-L1 antibodies, demonstrating how mechanistic understanding of IgG disposition (FcRn-mediated recycling, albumin effects) can be operationalized into regulatory-grade dosing recommendations.

Statistical Rigor Assessment

The analysis demonstrates strong statistical rigor appropriate for regulatory decision-making. Sample size ($n = 2,110$) is substantial for PopPK, providing robust parameter estimation with RSEs $\leq 12\%$ for key structural parameters and $\leq 30\%$ for covariate effects. The bootstrap validation ($1,000$ replicates) confirmed parameter stability with tight confidence intervals. Missing data rates were low ($< 1.5\%$ for most covariates), minimizing imputation bias. The sensitivity analysis framework—univariate covariate effects with fixed reference values, plus exploratory EBE-based subgroup analyses—appropriately distinguishes confirmatory from exploratory findings. The time-varying clearance model addresses a known mechanistic complexity (disease progression, TMDD) without over-parameterization, though the high shrinkage on Q (67.3%) and E_{max}

(46.7%) indicates some information limitation. The predefined 20% clinical relevance threshold for covariate effects aligns with regulatory guidance and was consistently applied.

⊠ Key Findings

The final PopPK model identified body weight (power 0.514 on CL, 0.470 on V_c) and albumin (power -0.714 on CL, -0.320 on V_c, -1.05 on V_p) as the most influential covariates, consistent with FcRn-mediated IgG disposition. Female sex reduced CL and V_c by $\sim 14\%$. Tumor type-specific baseline clearances ranged $0.171 - 0.211$ L/day (SCLC lowest, Other highest). Time-varying clearance decreased 9.3% from baseline with $T_{50} = 221$ days. Critically, flat-dose (200 mg Q2W, 300 mg Q3W) vs. weight-based (3 mg/kg Q2W, 4.5 mg/kg Q3W) simulations showed: (1) geometric mean exposure differences $\leq 6.59\%$ (well below FDA 20% threshold); (2) comparable exposure variability across body weight quartiles; (3) $C_{max,ss}$ with flat-dosing ($138 - 166$ $\mu g/mL$) far below 10 mg/kg Q2W reference (384 $\mu g/mL$); (4) efficacy predictions (OS, ORR) essentially identical ($\leq 0.1\%$ difference); (5) no exposure-safety relationships for any AE category. Even in lowest weight quartile ($33 - 55$ kg), 33.9% higher exposure with flat-dosing remained within safe bounds.

Clinical & Regulatory Implications

This analysis provides definitive quantitative support for transitioning serplulimab to flat-dose regimens (200 mg Q2W, 300 mg Q3W) as standard of care. The findings enable simplified prescribing without dose calculators, reduced pharmacy preparation errors, and improved supply chain efficiency. No dose adjustments are warranted for: body weight ($32.9 - 131$ kg range), sex, albumin variations, tumor burden, age, race, mild hepatic or renal impairment, or ADA positivity —all covariate effects were $< 20\%$ and clinically non-meaningful. The tumor type-specific parameter estimates allow indication-appropriate dosing while maintaining unified flat-dose strategy. For regulatory submissions, this work establishes a benchmark for PK comparability documentation and E-R integration supporting dose regimen changes.

Strengths & Limitations

Strengths

- Large, diverse dataset ($2,110$ patients, 11 trials, 6 tumor types) providing robust parameter estimation and broad generalizability
- Tumor type-specific modeling approach preserving indication-level applicability rather than naive pooling
- Rigorous time-varying clearance characterization addressing disease-related PK changes without over-parameterized TMDD
- Comprehensive simulation framework with explicit FDA-aligned comparability criteria (20% AUC/C_{min}, 25% C_{max} thresholds)
- Integrated efficacy and safety E-R analyses demonstrating maintained benefit-risk profile

- Appropriate handling of covariate effects with predefined clinical relevance thresholds
- Strong model diagnostics including pcVPCs stratified by tumor type and bootstrap validation

Limitations (Acknowledged by Authors)

- High eta shrinkage for Q (67.3%) and E_{max} (46.7%) indicating limited information for precise individual estimation of these parameters
- Moderate-to-high shrinkage on V_p (35.8%) suggesting peripheral compartment data may be sparse in some patients
- Proportional hazards assumption for OS E-R less well supported in ESCC, potentially reflecting delayed clinical benefit with immunotherapy
- Very limited data for moderate hepatic impairment ($n = 4$), precluding definitive conclusions
- ADA-positive patients only 6.3% of population, limiting immunogenicity impact assessment
- No dedicated organ impairment trials; hepatic/renal effects based on subgroup analysis of baseline lab values

Limitations (Expert Review)

- Lack of external validation dataset to confirm model predictive performance in independent patients
- No explicit model-based meta-analysis (MBMA) comparison with other PD-1 inhibitors to contextualize serplulimab PK within class
- Limited exploration of time-varying covariates (e.g., albumin changes during treatment) that could improve long-term exposure prediction
- No physiologically-based PK (PBPK) framework to mechanistically link covariate effects (particularly albumin) to FcRn recycling
- Simulation-based E-R for efficacy relies on previously developed exposure-response models not fully described in this manuscript; model structure and validation should be detailed for complete transparency
- No assessment of inter-occasion variability (IOV) for repeated dosing, which may be relevant given time-varying clearance and potential accumulation effects

Generalizability

Findings are highly generalizable to serplulimab's indicated populations (ES-SCLC, ESCC, Sq. NSCLC) and likely applicable to other PD-1/PD-L1 inhibitors with similar IgG4 backbone and FcRn-mediated disposition. The methodological framework—particularly the integration of PK comparability criteria with E-R confirmation—provides a template for flat-dose justification across oncology biologics. However, direct extrapolation to other antibody classes (e.g., ADCs, bispecifics) or non-oncology indications requires caution due to differing target biology and clearance mechanisms.

Key Equations

Baseline Clearance (CL0) with Covariates

$$CL_{0,i}\left(\frac{L}{day}\right) = CL_{TUMTP} \times \left(\frac{WT}{62}\right)^{0.514} \times \left(\frac{ALB}{41.4}\right)^{-0.714} \times \left(\frac{TUMBUR}{73}\right)^{0.0548} \times \left(\frac{ALP}{94}\right)^{0.0553} \times \exp[-0.145 \cdot (Female)] \times 24$$

Baseline clearance for individual i, incorporating allometric scaling on weight, albumin effect (inverse relationship), tumor burden, alkaline phosphatase, and sex effect (14.5% lower clearance in females). Tumor type-specific typical value CL_{TUMTP} with 24-hour conversion.

Time-Varying Clearance

$$CL_i = CL_{0,i} \times \exp\left[\frac{E_{max} \cdot \left(\frac{time}{T_{50}}\right)^{\lambda}}{1 + \left(\frac{time}{T_{50}}\right)^{\lambda}} + \eta_{CL,i}\right]$$

Sigmoid Emax model describing disease-related clearance changes over treatment time, with $T_{50} = 221$ days, $\lambda = 2.43$, and $E_{max} = -0.0926$ indicating 9.26% reduction from baseline. Inter-individual variability on clearance included.

Central Volume of Distribution (Vc)

$$V_{c,i}(L) = V_{c,TUMTP} \times \left(\frac{WT}{62}\right)^{0.470} \times \left(\frac{ALB}{41.4}\right)^{-0.320} \times \exp\left[-0.14 \cdot (Female) + \eta_{V_c,i}\right]$$

Tumor type-specific central volume with weight allometry, inverse albumin relationship, and 14% reduction in females. Inter-individual variability included.

Inter-compartmental Clearance (Q)

$$Q_i\left(\frac{L}{day}\right) = 0.0169 \times \exp\left(\eta_{Q,i}\right) \times 24$$

First-order inter-compartmental clearance between central and peripheral compartments, with only inter-individual variability (no covariate effects retained in final model).

Peripheral Volume of Distribution (Vp)

$$V_{p,i}(L) = 2.98 \times \left(\frac{ALB}{41.4}\right)^{-1.05} \times \left(\frac{TUMBUR}{73}\right)^{0.107} \times \exp\left(\eta_{V_p,i}\right)$$

Peripheral compartment volume with strong inverse albumin effect (power -1.05) and modest positive tumor burden effect. Fixed typical value with inter-individual variability.

Time-Varying Clearance Parameters

$$T_{50} = 221\ (day), \lambda = 2.43$$

Sigmoid Emax model parameters: T_{50} is time to 50% of maximum clearance change; λ (Hill coefficient) describes steepness of time effect.

Maximum Clearance Change (Emax)

$$E_{max,i} = -0.0926 + \eta_{E_{max},i}$$

Population mean maximum fractional change in clearance from baseline (-9.26% reduction) with inter-individual variability.

Figures & Tables

- **Figure 1**: Forest plots showing sensitivity analysis of significant covariates (body weight, albumin, sex, tumor burden, ALP, tumor type) on steady-state exposure metrics: $C_{avg,ss}$ (panel A), $C_{max,ss}$ (panel B), and $C_{min,ss}$ (panel C).
  - *Significance*: Demonstrates that all covariate effects on exposure are $\leq 20\%$ (clinical relevance threshold), with body weight and albumin having largest but still modest effects ($\sim \pm 20\%$). Critical visual evidence supporting no dose adjustment needed for any covariate.
- **Figure 2**: Predicted steady-state exposure ($C_{avg,ss}, C_{max,ss}, C_{min,ss}$) across exploratory covariate subgroups: age (≤ 65 vs > 65), race (Asian vs White), hepatic function (normal vs mild impairment), renal function (normal vs mild/moderate impairment), and ADA status.
  - *Significance*: Confirms minimal exposure differences in under-represented subgroups ($\leq 12\%$ for race, $\leq 7\%$ for hepatic, $\leq 22\%$ for renal), supporting unified flat-dosing across diverse patient populations. Note: moderate hepatic impairment ($n = 4$) excluded due to limited data.
- **Figure 3**: Comprehensive dosing regimen comparison: panels A-B show predicted C_{min1} distributions for 200 mg Q2W vs 3 mg/kg Q2W and 300 mg Q3W vs 4.5 mg/kg Q3W; panels C-D show C_{min1} vs body weight relationships; panel E shows concentration-time profiles for all five regimens including 10 mg/kg Q2W reference.
  - *Significance*: Core evidence for flat-dose justification: overlapping exposure distributions, comparable weight-exposure relationships, and flat-dose exposures well below 10 mg/kg Q2W safety ceiling. Visual demonstration of FDA comparability criteria satisfaction.
- **Figure 4**: Exposure-response analysis for efficacy: panels A-C show Kaplan-Meier OS curves comparing WT-based vs flat-dose predictions in Sq. NSCLC, SCLC, and ESCC; panel D shows ORR vs C_{min1} relationship in MSI patients; panel E shows predicted ORR comparison.
  - *Significance*: Demonstrates negligible efficacy differences ($\leq 0.1\%$ OS, overlapping ORR) between dosing strategies, confirming maintained clinical benefit with flat-dosing. E-R flatness supports exposure comparability as surrogate for outcome comparability.
- **Figure 5**: Safety exposure-response: probability of grade ≥ 3 TEAEs, G3TRAE, SAEs, AESIs, and irAEs across C_{max1} sextiles, with individual observations (blue circles) and observed rates with 95% CI (black circles with error bars).
  - *Significance*: Shows flat safety E-R relationships (no positive correlation between exposure and any AE type), supporting that higher exposures in low-weight patients with flat-dosing do not increase toxicity risk. Critical for benefit-risk assessment.
- **Table 1**: Patient characteristics and covariate distribution for the PK dataset ($n = 2,110$): demographics, laboratory values, tumor characteristics, and concomitant therapies.
  - *Significance*: Describes the diverse, representative population supporting broad generalizability. Key features: wide weight range ($33 - 131$ kg), predominantly Asian (87%), multiple tumor types, low ADA prevalence (6.3%), minimal missing data.

- **Table 2**: Final PopPK parameter estimates with RSEs, 95% CIs, and bootstrap validation: structural parameters (CL_0, V_c, Q, V_p), time-varying clearance parameters (E_{max}, T_{50}, λ), covariate effects, and variability components (IIV, residual error).
  - *Significance*: Core model characterization with excellent precision (RSE $\leq 12\%$ for key parameters, $\leq 30\%$ for covariates). Tumor type-specific CL_0 and V_c values shown. Bootstrap confirms parameter stability. Eta shrinkage values indicate information content for random effects.

Code & Reproducibility Assessment

Analysis performed in NONMEM (version not specified, likely $7.x$) with standard PopPK methodology. No code or model files provided in supplementary materials—only a data supplement (psp470204-sup-0001-datas1.docx) of unspecified content. Reproducibility limited to methodological description; no executable code, control streams, or datasets available for independent verification. Standard reporting of diagnostic plots and VPCs suggests conventional workflow but precludes direct replication.

Supplementary Materials

One supplementary file provided: psp470204-sup-0001-datas1.docx (description not detailed in available content). Likely contains additional data tables, possibly extended patient characteristics, or trial-level summaries. No supplementary figures, model code, or additional methodological details referenced in main text extract.

Future Directions

Key questions emerging from this work include: (1) Can the time-varying clearance model be refined with longitudinal albumin/tumor burden data to improve long-term exposure prediction? (2) Would a dedicated hepatic impairment trial (Child-Pugh B/C) support dosing in populations currently under-represented? (3) Can PBPK modeling mechanistically link the observed albumin-covariate relationship to FcRn saturation kinetics? (4) How does serplulimab's PK-E-R relationship compare quantitatively with pembrolizumab, nivolumab, and other anti-PD-1 agents via MBMA? (5) Would adaptive trial designs incorporating real-time PK monitoring validate the flat-dose assumption in special populations? (6) Can the tumor type-specific parameter estimates inform combination therapy optimization (e.g., chemotherapy-PK interactions)? The model structure also invites extension to QSP frameworks linking serplulimab exposure to receptor occupancy and downstream immune cell dynamics for mechanistic dose optimization.

Expert Commentary

This manuscript exemplifies the evolution of pharmacometrics from descriptive characterization to strategic drug development decision-making. The authors' approach—eschewing TMDD for empirical time-varying clearance—demonstrates pragmatic model selection when mechanistic complexity exceeds information content, a lesson for PopPK practitioners often seduced by bio-

logical realism. The tumor type-specific parameterization, while increasing model dimensionality, preserves clinical utility for indication-specific dosing recommendations and reflects growing regulatory expectation for subgroup-relevant PK documentation. Most importantly, the seamless integration of PK comparability simulations with E-R confirmation illustrates the 'confirmatory pharmacometrics' paradigm now essential for post-approval dose optimization. The 20% clinical relevance threshold, applied consistently across covariates and dosing comparisons, provides a transparent decision framework that should be adopted more broadly. For the field, this work reinforces that flat-dosing justification requires more than overlapping concentration distributions —it demands demonstrated equivalence in efficacy and safety outcomes, achieved here through rigorous E-R modeling. The high shrinkage on inter-compartmental parameters reminds us that even large datasets may not resolve all structural uncertainty, and that parameter precision (RSE) must be weighed against identifiability. Finally, the manuscript's alignment with FDA's 2022 PD-1/PD-L1 dosing guidance demonstrates the value of proactive regulatory engagement in pharmacometric analysis planning.

Bottom Line

For practicing pharmacometricians, this paper provides a regulatory-grade template for justifying dose regimen transitions, particularly the shift from weight-based to flat-dosing for monoclonal antibodies. Key takeaways: (1) Predefine clinical relevance thresholds (here 20%) before covariate analysis to avoid post-hoc rationalization; (2) Time-varying clearance may be more identifiable than full TMDD when target saturation is rapid; (3) Tumor type-specific parameters balance model parsimony with clinical applicability; (4) PK comparability alone is insufficient—E-R confirmation for efficacy AND safety is essential; (5) Bootstrap + pcVPC stratified by key subgroups provides robust validation; (6) Document that even 'worst-case' exposure scenarios (e.g., low weight + flat dose) remain within established safety margins. The serplulimab case demonstrates that with adequate data ($n > 2000$, diverse indications), rigorous PopPK can drive operational simplification without compromising patient outcomes—a benchmark for future antibody dose optimization programs.

⊠ Figures

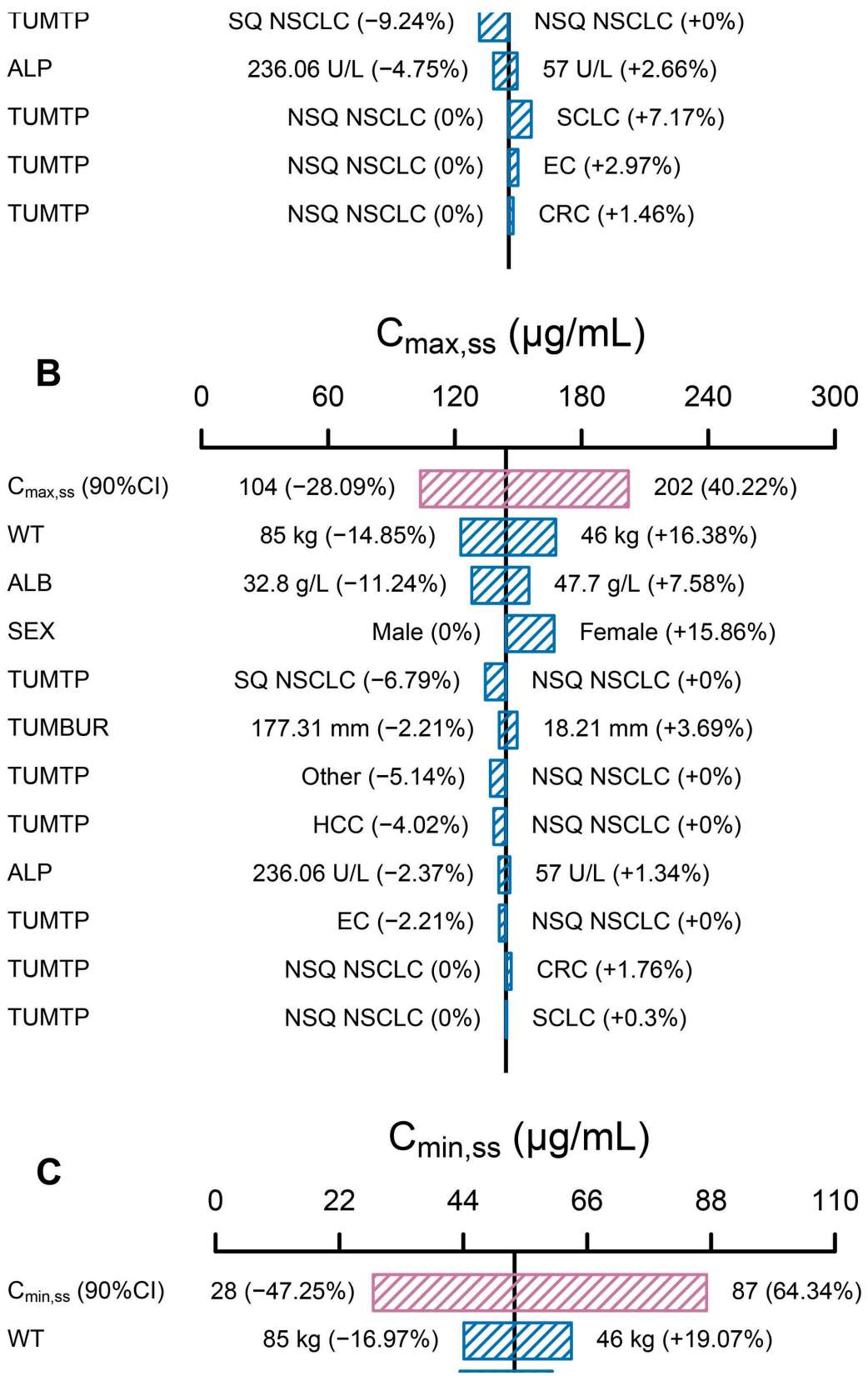

Figure 1: FIGURE 1: Sensitivity analysis showing the influence of significant covariates on steady-state exposure (A)Cavg,ss, (B)Cmax,ss, and (C)Cmin,ss. The vertical black line r

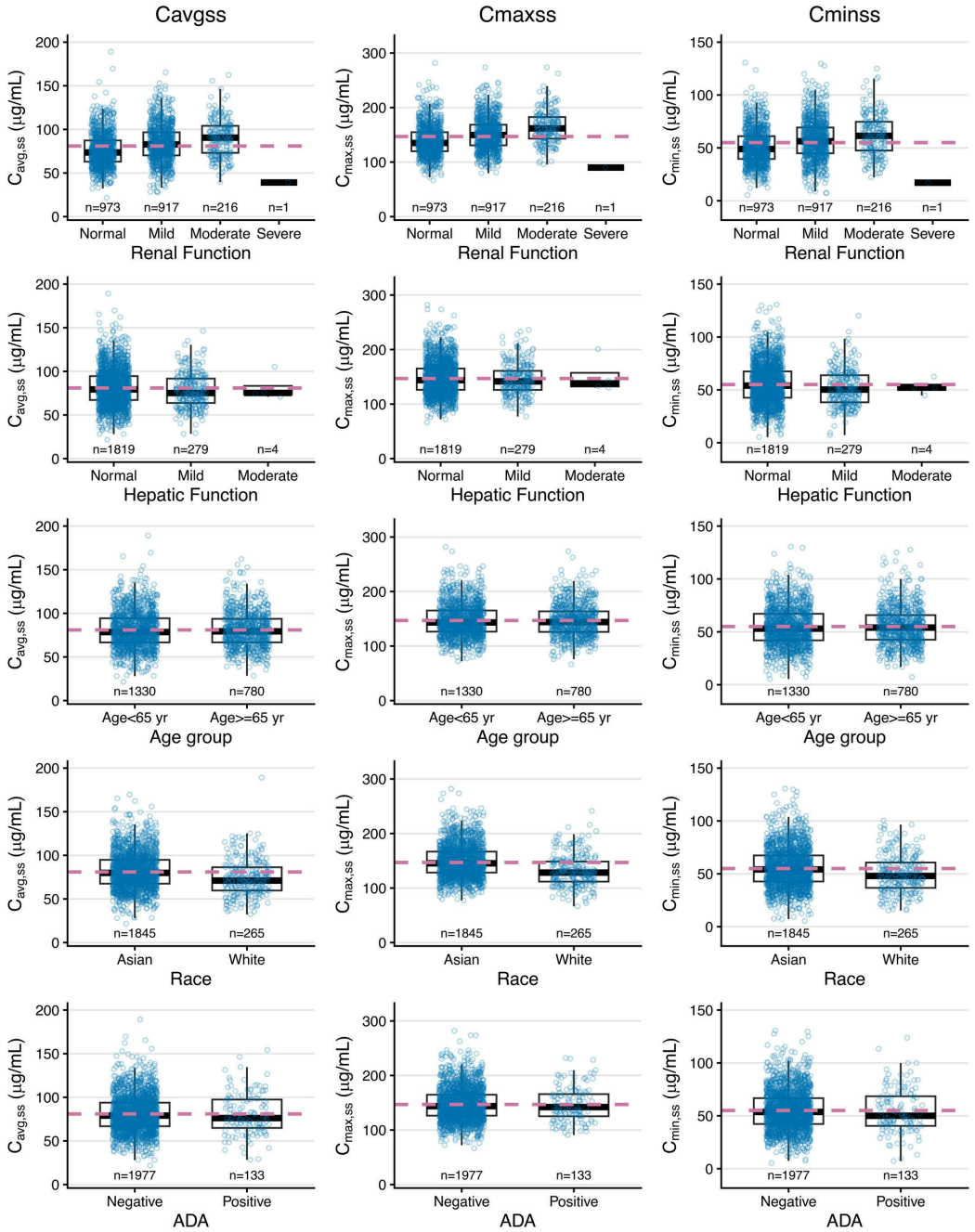

Figure 2: FIGURE 2: Predicted steady-state exposure (Cavg,ss,Cmax,ss, andCmin,ss) across key covariates subgroups. Model-predicted exposure metrics were summarized by renal function

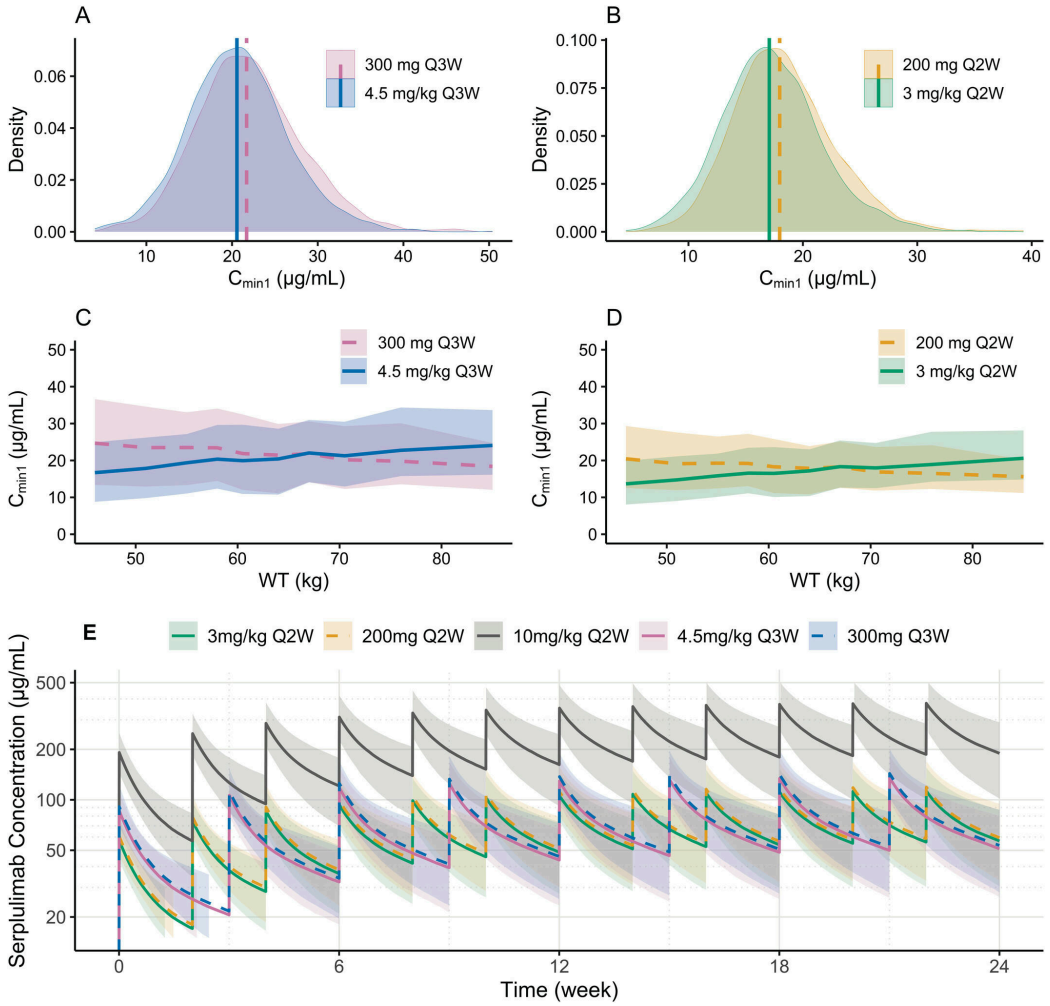

Figure 3: FIGURE 3: Predicted trough concentrations (Cmin1) and time–concentration profiles following different serplulimab dosing regimens. Panels (A, B) show the predicted distrib

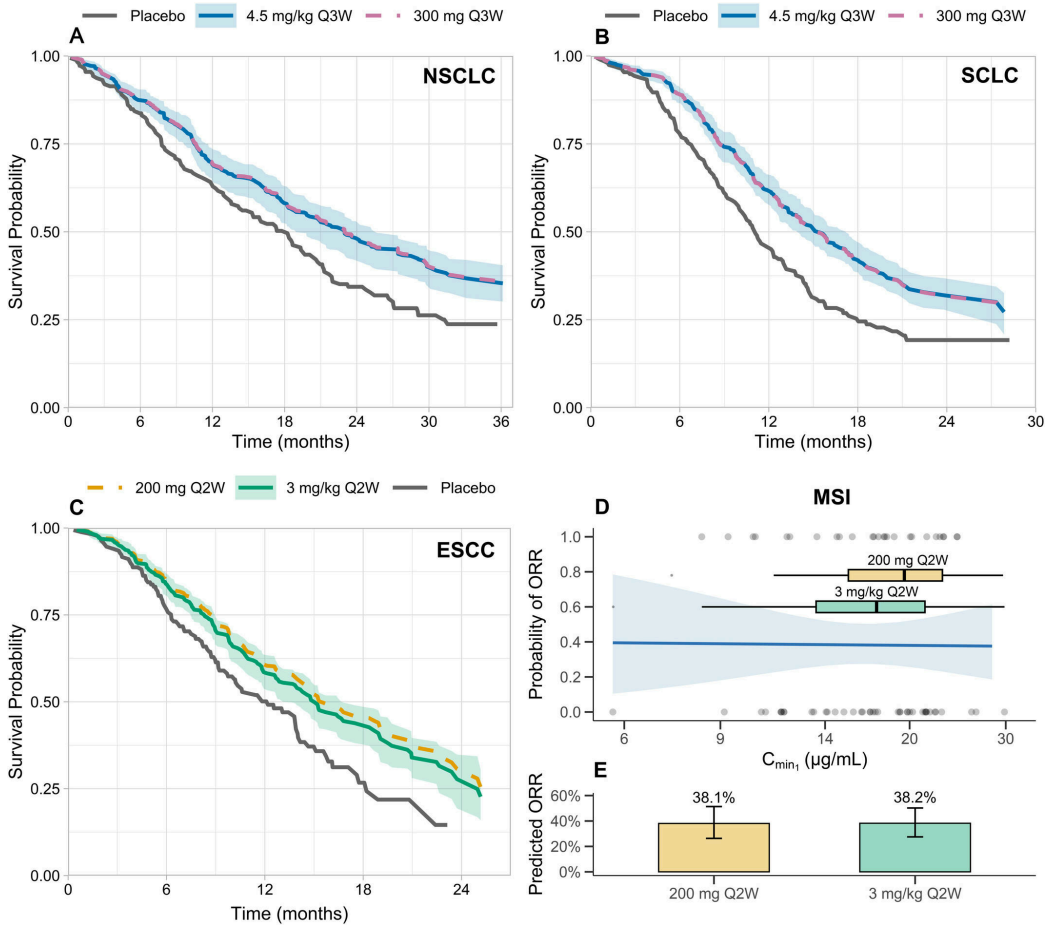

Figure 4: FIGURE 4: Exposure–response (E–R) analysis for efficacy across tumor types and dosing regimens. Kaplan–Meier plots show overall survival for control and serplulimab treatm

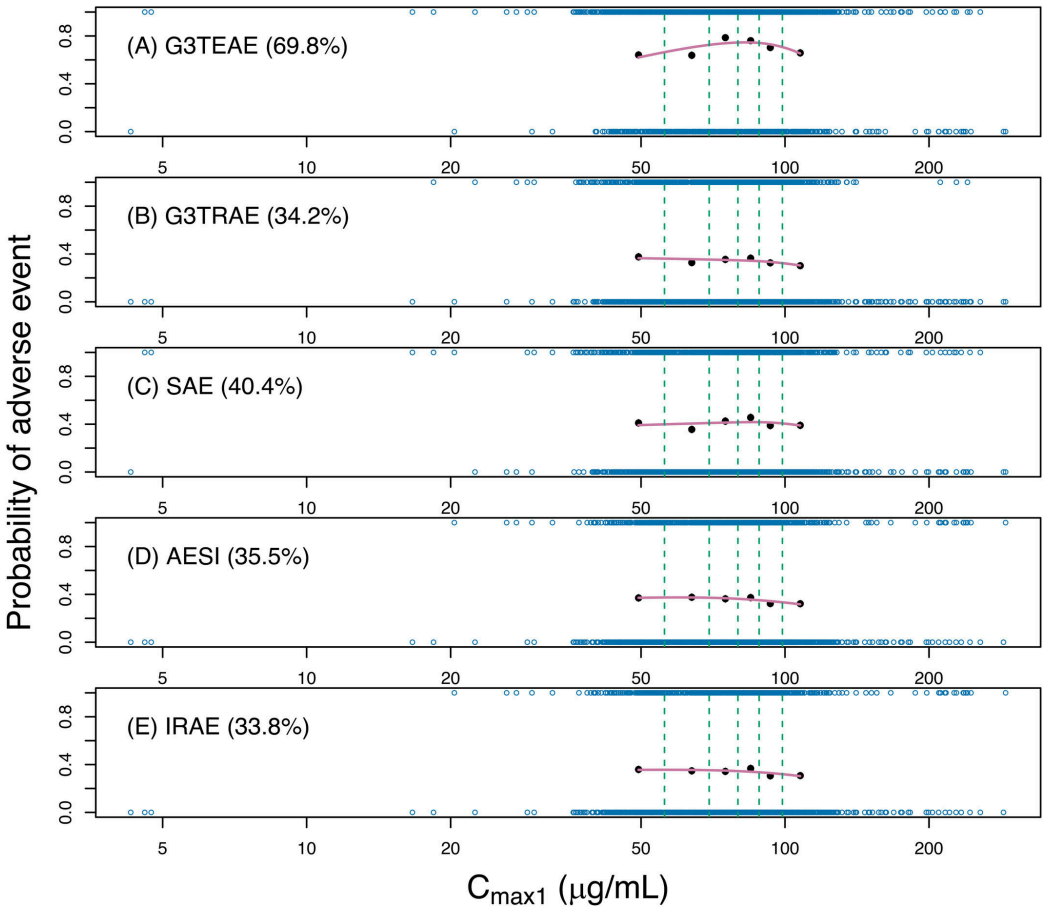

Figure 5: FIGURE 5: Probability of adverse events (AEs) across predicted serplulimab Cmax1sextiles. Blue open circles represent individual observations; black solid circles show obs